Science Homework: **Work & Power** **Due on:**

Name: _____ Mark: _____/30

$E_p = mgh$; $W = F \times d$; $P = \frac{W}{t}$; $g = 10 Nkg^{-1}$

1) Complete the following table:

Quantity	Letter	Unit	Abbreviation
Mass	m	kilogram	kg
Work			
	P		
	F		N
		kilogram metre per second	$kgms^{-1}$

(10 marks)

2) What is the work done when:
   a) A car is pushed 450m by a force of 900N? _____
   _____(_____)
   (2 marks)
   b) A bullet is pushed by a force of 1.2kN for 1.8km?
   _____
   _____(_____)
   (2 marks)

3)
   a) How far can you push a cart if the force is 340N and the work done is 2.5kJ?
   _____
   _____(_____)
   (2 marks)
   b) What force is required if 1200J of work moves a box 34m?
   _____
   _____(_____)
   (2 marks)

4)
   a) What is the power output if 4500J of work is done in 1.5s?
   _____
   _____(_____)
   (2 marks)
   b) How much energy is required for a 120W light to shine for 3 minutes?
   _____
   _____(_____)
   (2 marks)

5) A man lifts a 23kg bag 1.4m onto a table in 1.6s. Calculate:

a) The work done

_____

_____(_____)

(2 marks)

b) The power output

_____

_____(_____)

(2 marks)

6) A feather is dropped on a calm day, using your knowledge of forces, acceleration and velocity describe its motion as fully as possible. A sketch may help in explaining.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

(4 marks)